DISRUPTION_X

June 2025

America's nuclear renaissance has begun

"I want one."

That's the gut reaction Isaiah Taylor wants to elicit when you first see a Valar Atomics nuclear reactor.

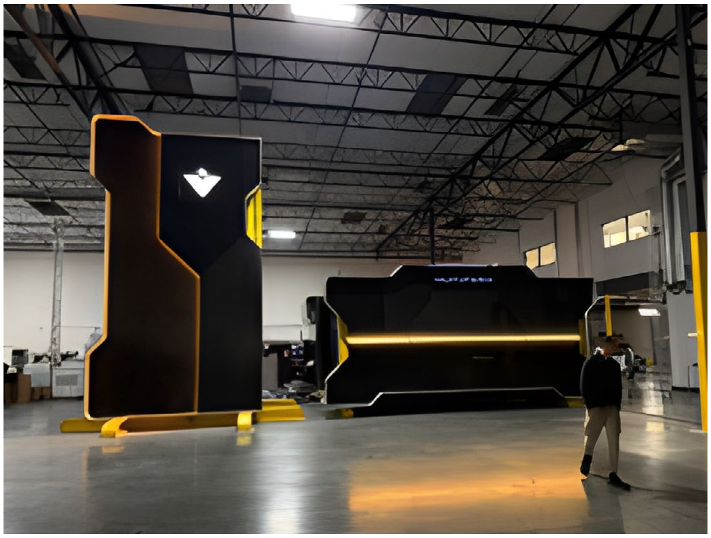
RiskHedge publisher Dan Steinhart and I visited Valar's HQ, located about 10 minutes south of LAX airport, back in April.

When I think of nuclear power plants, I envision giant concrete buildings and old Soviet-style control rooms. Here's "old" nuclear:

Source: The Wall Street Journal

In this month's issue

Innovators like Valar Atomics are ditching this drab image. Check out Valar's sleek nuclear rig reactor and control room:



Inside at Valar Atomics

Dan and I stood inside one of Valar's control rooms. It felt like a high-end gamer studio. Nuclear should look and feel like the future. Now, it finally does.

For the seven years I've written about nuclear, the story has been one of grim survival. The investment case was, "I ran the numbers, and it looks like we won't shut down as many plants as expected."

As for innovation, to quote the great philosopher Tony Soprano, "Fuhgeddaboudit."

Now, the world is making a 180-degree turn on atomic energy. Few investors grasp the speed of this shift. But after decades of stagnation, we're entering a nuclear renaissance.

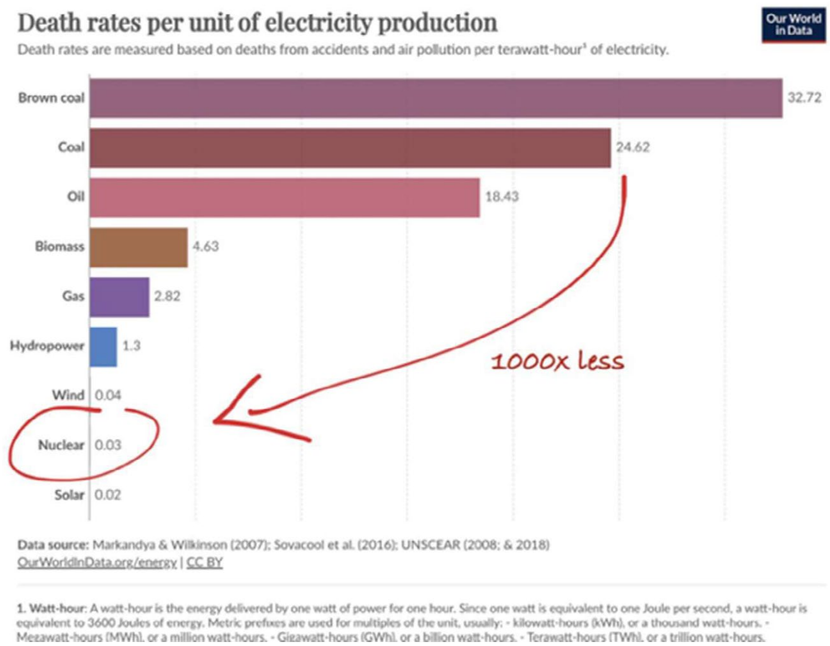
Chris will walk you through the most promising ways to invest in this trend below. First, let me show you how and why this revolution is unfolding.

Let's get the facts straight

Many folks' views of nuclear energy are shaped by watching *The Simpsons*, where Homer nods off in the control room and toxic green sludge leaks into local rivers.

The reality is that atomic energy is the cleanest, safest, densest, best energy source known to man. Let's walk through each point below.

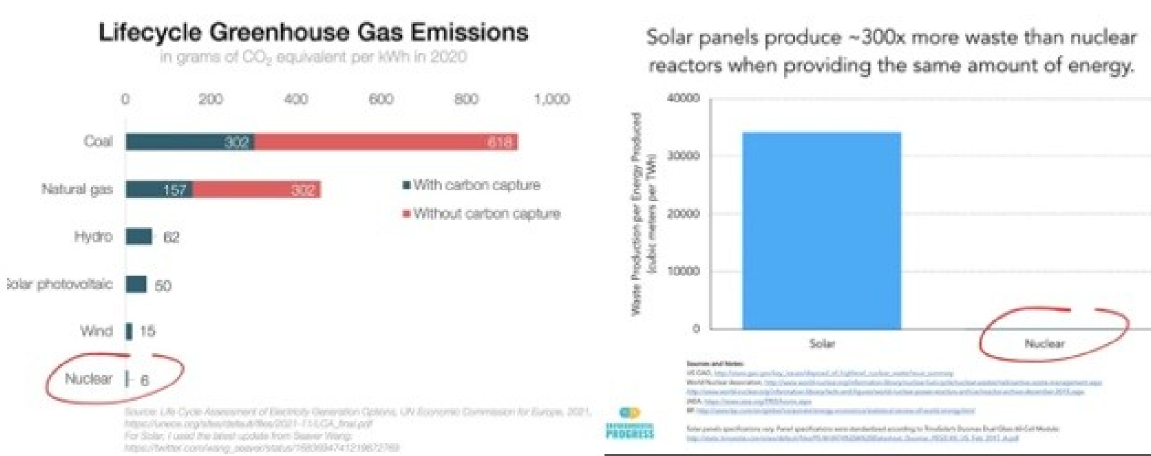
Safest: For every unit of energy produced, brown coal kills over 1,000X more people than nuclear. It's just as safe as solar. In fact, you receive more radiation from eating a single banana than you do from living next to a nuclear power plant for a year.



Source: Our World in Data

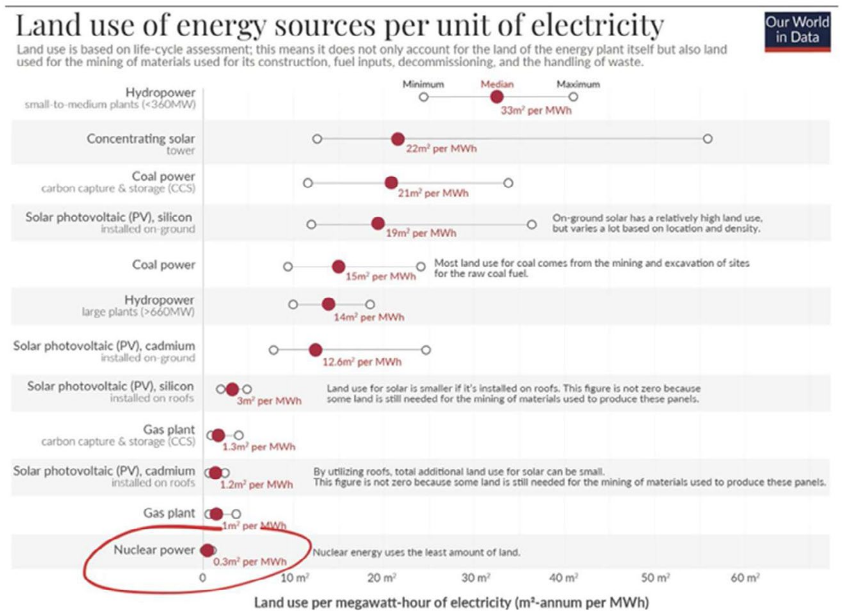
Cleanest: When accounting for the entire lifecycle, from mining to waste disposal, nuclear power emits less CO2 than any other source. And its waste is minuscule. All the US's nuclear waste produced over the past 60+ years could fit on a single football field, stacked less than 10 yards deep.

Solar panels, by contrast, produce 300X more waste by volume for the same amount of energy.



Sources: United Nations Economic Commission for Europe; Environmental Progress

Densest: A nuclear plant requires a tiny fraction of the land needed for solar or wind farms to produce the same round-the-clock power. If you love pristine, green hills, you should champion nuclear energy. Criticizing nuclear on environmental grounds is backward.

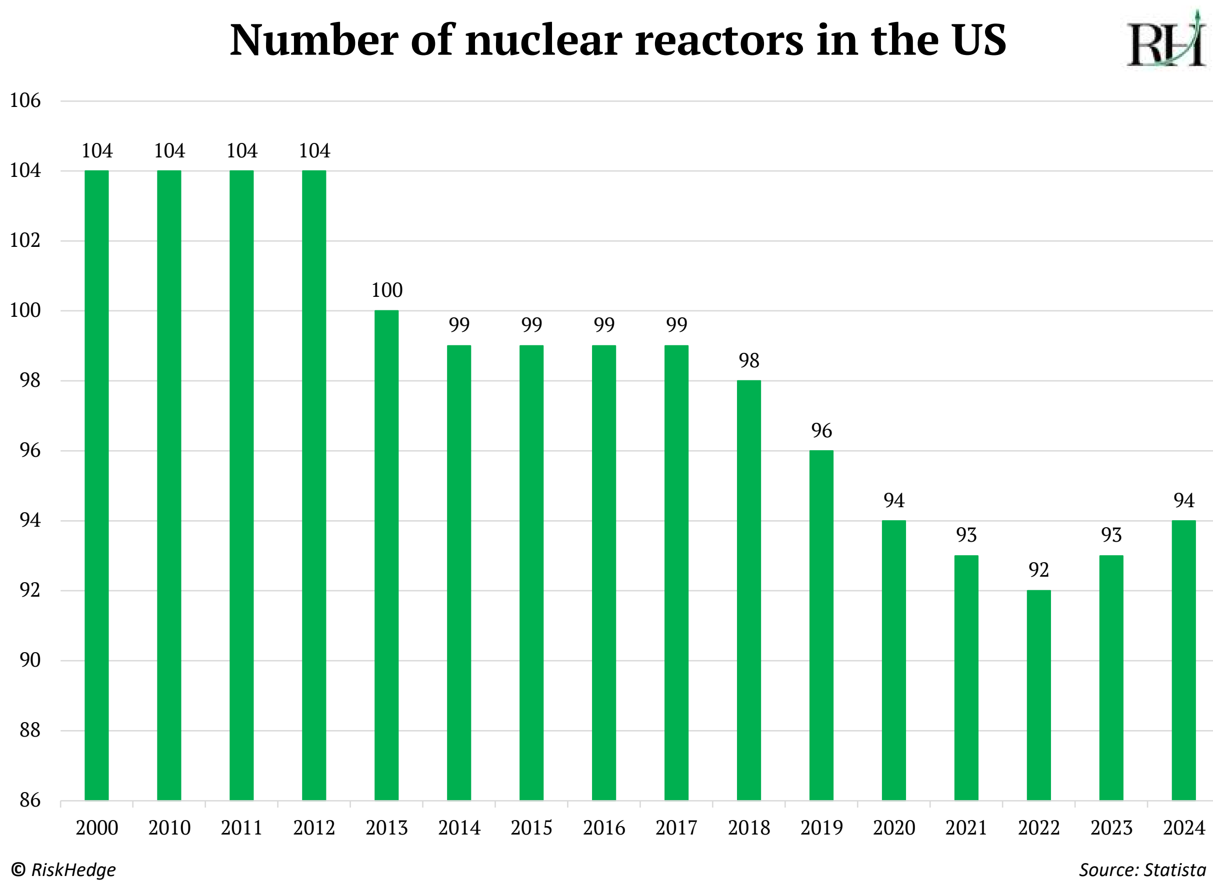


Source: Our World in Data

Nuclear energy is so safe, clean, and reliable that if aliens landed on Earth, they'd be baffled as to why we don't have atomic-powered everything.

My years as a professional investor have taught me narratives often rule over facts, at least in the short term. Unfortunately, for the past 50 years, the narrative of nuclear plants pumping out radiation and causing cancer has overruled the facts.

America has shut more nuclear reactors than we've built this century. Nuclear was an industry in decline until recently:



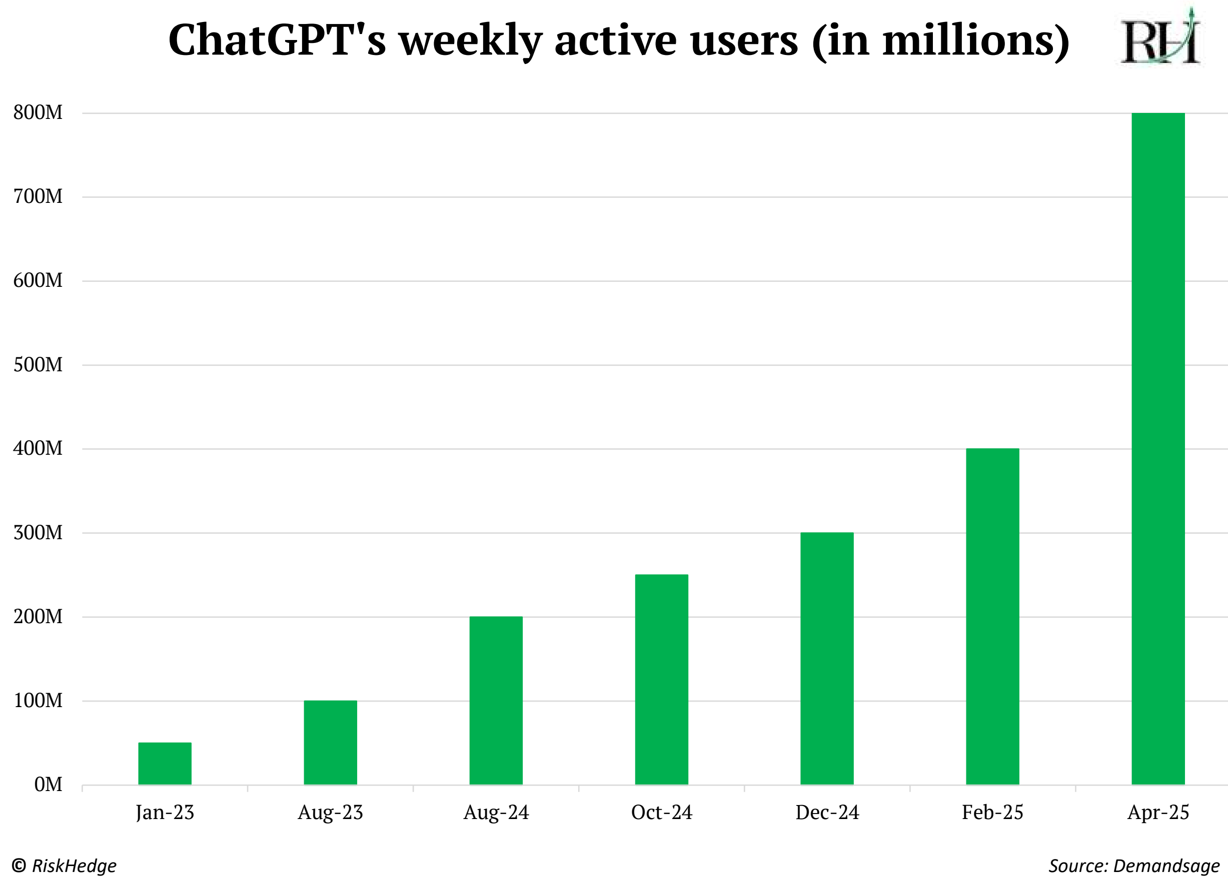
That is, until ChatGPT came along.

How ChatGPT sparked an energy revolution

When you use ChatGPT, you're tapping into vast data centers filled with specialized chips that run so hot, they need constant cooling just to keep from melting down. One of **Nvidia's (NVDA)** new artificial intelligence (AI) chips draws as much power as two households.

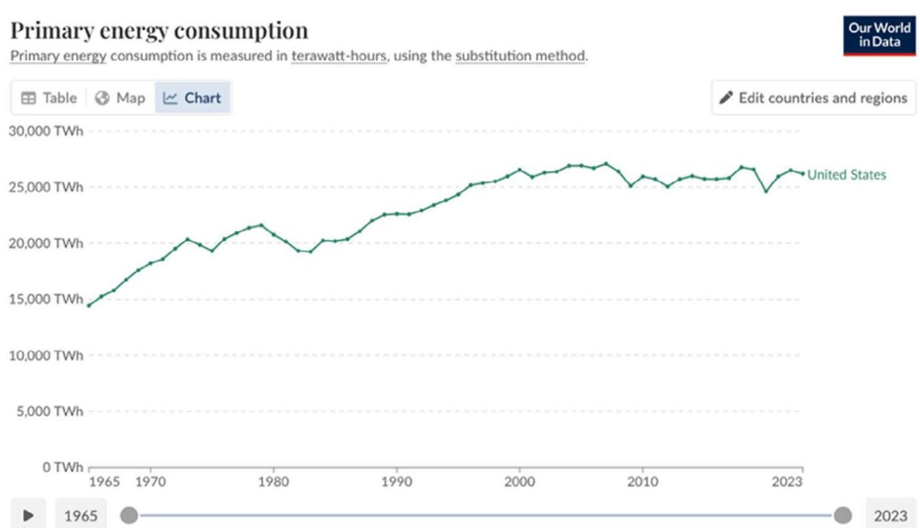
A modern AI data center needs 5X more power than a traditional one. Even a simple ChatGPT query uses *at least* 10X more electricity than a Google search.

ChatGPT launched almost three years ago. It's become the fastest-growing product in history, with roughly 800 million weekly users:



Now, everyone is asking, "How are we going to power all these AI factories?"

Ditching nuclear was a luxury. America's energy consumption has been in a 20-year bear market. "Why do we even need nuclear?" was a legitimate question.



Source: Our World in Data

Now with AI, electric cars, and robotics going gangbusters, we need all the cheap, clean energy we can get our hands on.

The CEO of power provider **NRG Energy (NRG)** recently said, "For the first time in... perhaps in my 40 years in this business, we are now expecting a step change in long-term power demand."

That, my friends, is forcing everyone to do a U-turn on atomic energy.

Two weeks ago, the World Bank lifted its ban on funding nuclear power projects. The ban has "officially" been in place since 2013. But the last time the bank funded a nuclear project was 1959. This is HUGE!

Nuclear plants are expensive to build but cheap to operate. Financing their construction is the key to having many more power plants.

Germany abandoning its world-leading nuclear fleet is one of the greatest own-goals of modern history. Shutting nuclear plants is basically energy suicide.

Electricity prices in Germany spiked, forcing many factories to shut down. Then it had to buy "dirty" oil and gas from Russia (and elsewhere) to keep the lights on. It was the poster child for a world gone mad.

Thankfully, Germany didn't completely go off the deep end and is now doing a 180.

Germany's new leader Friedrich Merz called exiting nuclear energy a mistake and said it will no longer object to treating nuclear power on par with renewable energy. Let's hope they turn those reactors back on!

Elsewhere in Europe...

- UK Prime Minister Keir Starmer unveiled plans for a historic expansion in nuclear power. He called for tech companies to work alongside the government to build small modular reactors (SMRs) to power AI data centers.
- Sweden plans to build an additional 5 GW of nuclear capacity by 2035.
- Denmark repealed a 40-year ban on nuclear reactors.
- Belgium voted to abandon its nuclear phase-out plan.

If you had read these headlines to me a few years ago, I wouldn't have believed you. It's amazing how a game-changing tech like AI can shift the narrative.

But the most important U-turn of all happened in the home of the world's nuclear superpower: the USA.

America still produces more atomic energy than any other country. But we've only opened two new power plants in the past 30 years.

To steal a line from Democratic strategist James Carville, “It’s the regulations, stupid.”

America was building nuclear reactors by the dozen until the early '70s. But in 1974, a bureaucratic creature named the Nuclear Regulatory Commission (NRC) was hatched. Guess how many new reactor designs it’s greenlit since taking control of nuclear 50 years ago?

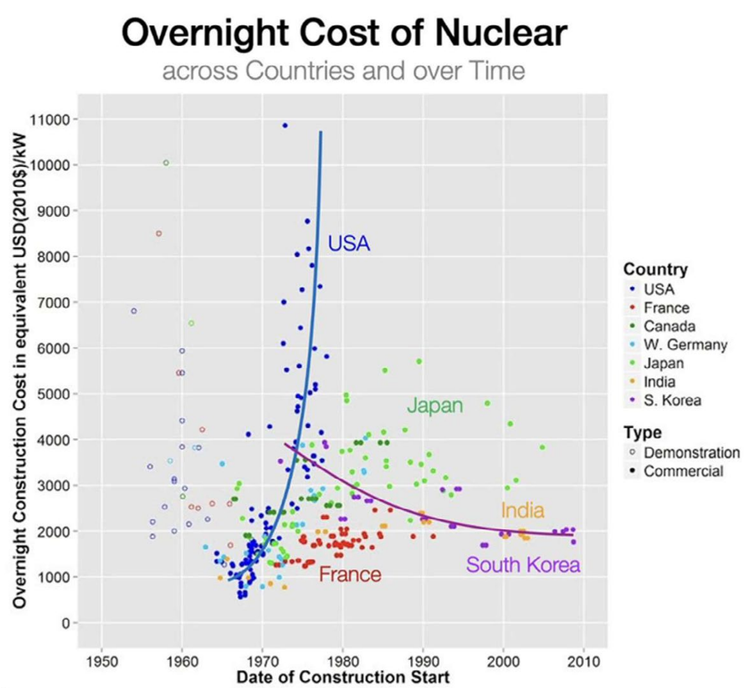
Zero!

The NRC didn’t exactly ban new nuclear plants, but it may as well have. Building a new reactor requires over 100,000 pages of documentation, takes 15 years, and costs tens of billions of dollars.

Nuclear companies must pay the NRC $300/hour for the privilege of that review. Talk about perverse incentives.

Turning our backs on nuclear was the greatest self-inflicted wound in American history. It’s why we’re still fighting about climate change when we could be enjoying abundant, clean, cheap energy.

In the '70s, producing electricity from nuclear cost about one-third of what it costs today. The tech didn’t change. In fact, it got even better. What changed were the rules.



Source: Tomas Pueyo on Substack

But I bring good news, friends.

Last month, President Trump signed four executive orders aimed at freeing nuclear energy from its regulatory shackles. We combed through the executive orders so you didn’t have to.

Here are the five big changes…

1. Set a goal of quadrupling the size of America's nuclear fleet by 2050.

2. Speed up the development of "advanced nuclear" (read: SMRs) through pilot programs and streamlining environmental reviews. This orders the NRC to license new reactors within 18 months.

3. Orders the Department of Energy to approve at least three reactors by mid-2026. Basically, Trump wants three SMRs up and running for America's 250th birthday.

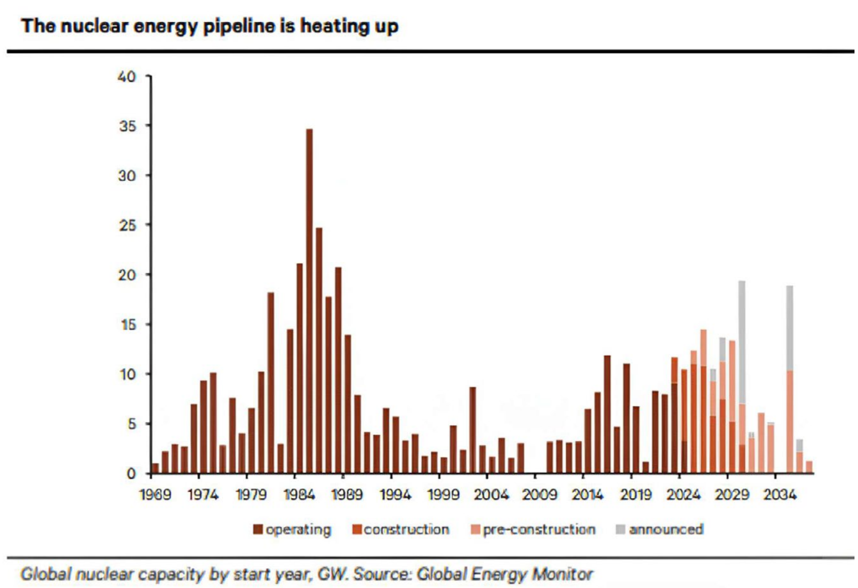
4. Designates nuclear plants that power AI facilities as "defense-critical infrastructure." Building nuclear-powered data centers on military bases is a genius loophole. It potentially allows projects to avoid lengthy NRC reviews.

5. Lastly, but in our view most importantly, asks the NRC to reconsider its "As Low As Reasonably Achievable" (ALARA) regulation. As mentioned, you receive more radiation from eating a single banana than you do from living next to a nuclear power plant for a year. Yet under the ALARA rule, even that isn't safe enough!

   This rule has made it basically illegal to build nuclear plants in America. I think Trump should have gone further and ordered the NRC to eliminate the rule. But we're moving in the right direction.

Isaiah Taylor said these changes mean Valar could have a reactor that's operational by *next year*. Another nuclear founder told me these orders are a "game-changer." Green light… go!

I certainly don't agree with Trump on everything. But if he can revitalize America's nuclear program, it'll be one of the greatest presidential achievements of the past century. If this succeeds, we all win.



Source: Global Energy Monitor

It’s clear we’re past peak nuclear pessimism. The chart above shows the nuclear pipeline is heating up—and it doesn’t even include all the SMRs that will be built.

Get ready for the second atomic age.

Nuclear is no longer a 1-customer industry

Previously, power companies were the sole buyers for nuclear energy—and they’re the definition of risk-averse. They tend to be slow and stuck in their ways. They certainly aren’t the first to adopt new technologies like SMRs.

This is why AI is nuclear’s best friend. AI data centers have an urgent need for massive amounts of clean, reliable energy. Nuclear provides that.

Why else do you think big tech companies are sprinting around, checkbooks in hand, hoovering up all the atomic energy they can get their hands on?

For example, in the past year:

- **Microsoft (MSFT)** inked a deal to help restart a reactor at the iconic Three Mile Island plant in Pennsylvania.

- **Google (GOOGL)** signed an agreement with startup Kairos Power to build several SMRs to power its data centers.

- **Amazon (AMZN)** entered a 17-year contract with **Talen Energy Corp. (TLN)** to power its data centers with nuclear energy. It’s also among a consortium investing $500 million into SMR startup X-energy.

- **Meta Platforms (META)** inked a 20-year deal to buy power from **Constellation Energy Corp.’s (CEG)** Clinton nuclear plant in Illinois. The deal ensures the Illinois plant will stay operational past its previously scheduled 2027 closing.

A few years ago, I wrote we’d soon seen nuclear-powered AI factories. It sounded a little kooky at the time. Now, it’s happening!

While most investors are racing to get up to speed on nuclear, we have a head start.

Chris and I have been writing about and investing in nuclear energy for years. We’ve also chatted with several key players in the industry, from entrepreneurs building SMRs to fuel suppliers.

That will pay dividends as the nuclear renaissance heats up.

The two biggest barriers to nuclear progress—a lack of customers and a hostile regulatory environment—are crumbling.

The third piece of the puzzle is technology. Forget the huge, grey, ugly cooling towers. Our atomic future is small and beautiful, and it's being forged by innovators building reactors with their own hands across America.

The old model of building reactors as one-off, bespoke "cathedrals" is also dying. In its place, a new vision is rising: the nuclear gigafactory.

Companies are now designing microreactors that can be mass-produced on an assembly line, just like a jet engine or a car.

Valar Atomics plans to put hundreds of identical small reactors clustered on a single "Gigasite." Not one behemoth, but an army of identical twins working in harmony.

In Austin, Texas, another startup we visited called **Aalo Atomics** is building a factory with the goal of having it ship standardized nuclear "pods" 60 days after an order is placed.

In California, **Radiant Industries** is developing microreactors that fit inside a shipping container and can replace diesel generators in remote locations.

We're swapping one-off cathedrals for repeatable Camrys, making nuclear better… faster… and cheaper in the process.

As excited as I am about the nuclear renaissance, I must put my investor hat on and stay sober. There's a right way and a wrong way to invest in nuclear.

I'll hand you off to Chris to talk about the opportunities—and pitfalls—of investing in the nuclear renaissance.

Mapping the nuclear value chain

Before I (Chris Wood) invest in an emerging megatrend, I like to make a mental map of its underlying ecosystem.

This lets me develop a working knowledge of the different kinds of companies enabling the trend and driving it forward. In turn, this helps me quickly identify potential opportunities to capitalize on.

To make this mental map, I examine the trend through the lens of its "value chain."

A "value chain" is simply the set of steps, phases, or stages that add value to a product during its journey of being made.

Think about your favorite pizza joint, for example. Before it serves you a hot, delicious pizza, there are several steps required, each of which adds value to the final product.

First, farmers must grow wheat for flour and raise cows to make the milk for cheese.

Next, manufacturers turn these raw ingredients into flour, cheese, and tomato sauce—adding more value by making them readily usable.

Then distributors get the ingredients to your local pizza place, adding value by making them available where and when they're needed.

Finally, your pizza restaurant works its magic. It combines the ingredients, adds its special secret touches, bakes it in its special ovens, and serves it to you hot and fresh.

Each step along the way—from farm to table—makes the pizza more valuable than it was before.

Every product or service goes through similar stages where value gets added. So every trend can be viewed through this "value chain" lens.

Understanding a megatrend's value chain helps us:

- Identify its various opportunities and challenges.
- Figure out how different companies fit into its ecosystem.
- Home in on its biggest potential beneficiaries.

Today, we're concerned with the emerging nuclear megatrend and its value chain.

Nuclear power comes from splitting atoms of heavy elements in a process called fission. This process releases a huge amount of energy as heat, which is used to make steam and drive turbines to produce electricity.

The nuclear power value chain is the series of stages necessary to turn nuclear fuel into usable electricity and safely dispose of the waste.

I'll map it out for you from a high-level to give you an idea of how we're approaching the space. Then, we'll drill down on one specific layer of the value chain and talk about some potential emerging investment opportunities in it.

The four broad layers I came up with for the nuclear power value chain are:

1. The foundation layer: Sourcing and preparing nuclear fuel.
2. The power generation layer: Developing and operating nuclear reactors.
3. The support layer: Maintaining and servicing nuclear power plants.
4. The cleanup layer: Managing and disposing of nuclear waste.

Let's walk through each layer below.

The foundation layer: Sourcing and preparing nuclear fuel

Every nuclear power plant needs fuel, just like a car needs gas and you need food.

The foundation layer is all about acquiring and preparing the fuel that nuclear reactors use—mainly uranium (and new fuels in the future). The journey from raw uranium ore to reactor-ready fuel is the first layer of the value chain. It's like gathering and prepping your ingredients before cooking a meal.

Without fuel, there's no nuclear power. So this is the foundation. Some key steps and innovations here include:

Uranium mining

Uranium is a heavy element found in certain rocks. Mining companies like **Cameco Corp. (CCJ)** dig it out of the ground in places like Canada, Kazakhstan, Australia, Namibia, Niger, and Russia.

Over 85% of the world's uranium comes from these six countries. The raw ore is crushed and chemically processed into a concentrated powder called yellowcake, which is like the "flour" that will be refined into nuclear fuel.

Enrichment

Companie like Urenco in Europe and TENEX in Russia then enrich the yellowcake, which means increasing the amount of a special type of uranium (a key isotope) it contains to fuel the nuclear reaction.

Fuel fabrication

Companies like Westinghouse and Framatome press the enriched uranium powder into small cylindrical pieces and bake them at a high temperature until they become hard ceramic pellets.

Each of these gummy-bear sized "fuel pellets" produces about as much energy as an entire metric ton of coal.

The pellets are loaded into long metal tubes made of special alloys (often zirconium) to create "fuel rods." Dozens to hundreds of rods are bundled into "fuel assemblies" that go into the reactor core.

Think of it like bundling a bunch of batteries into packs. The pellets are the "cells," and the assemblies are the "packs" that power the reactor.

Advanced fuels (innovations)

Researchers are developing new nuclear fuels and materials to make reactors safer and more efficient.

For example, TRISO fuel pellets—tiny uranium fuel kernels about the size of poppy seeds coated with layers of carbon and ceramic to lock in radioactive material—can stand extremely high temperatures (making them meltdown-proof) and act like their own containment vessels.

Another example is HALEU (high-assay low-enriched uranium), which is being developed by companies like **Centrus Energy Corp. (LEU)** and **ASP Isotopes (ASPI)** and contains a higher-than-"normal" amount of the key uranium isotope needed to fuel the nuclear reaction.

"Normal" in this context refers to what's called low-enriched uranium (LEU), which is what fuels most reactors.

Many of the advanced reactors being developed today, like SMRs, will use HALEU fuel because it lets smaller reactors run longer without refueling and get more power from a smaller core.

In short, the foundation layer takes us from rocks in the ground to high-tech fuel assemblies ready to power a nuclear reactor. It's capital intensive—involving heavy mining operations and sophisticated enrichment and fabrication facilities—but creates the essential fuel the entire value chain relies on.

The power generation layer: Developing and operating nuclear reactors

Once we have fuel, the next layer involves using it to generate power.

This second layer is where the magic of nuclear fission happens—splitting atoms to release huge amounts of energy. It's the heart of the value chain.

Companies like Constellation Energy, **Exelon (EXC)**, and **NextEra Energy (NEE)** operate large conventional nuclear reactors to generate power. These reactors are like giant, high-tech tea kettles that use nuclear fuel to boil water into steam, which then spin turbines to generate electricity.

It's just like how a coal-fired power plant works except the heat comes from nuclear fission instead of burning coal.

Over 400 such conventional reactors operate globally today. They generate roughly 9%–10% of the world's electricity. And while these behemoths—about two-thirds of which are over 30 years old—still dominate the nuclear power industry, there's a ton of research and development happening in this layer.

For example, you may have heard of the SMRs being developed by **NuScale Power Corp. (SMR)**, **Oklo (OKLO)**, and others.

NuScale's SMRs shrink traditional large-scale nuclear reactor tech down to about the size of a tennis court. They look less like a traditional power plant with giant concrete cooling towers and more like an oil tank.

SMRs have great potential because:

- They'll be a lot cheaper, easier to cool, and faster to build than conventional reactors.
- They'll make nuclear power available to more places in remote regions with smaller grids because of their significantly reduced footprint.
- They'll be even safer thanks to simplified designs with fewer moving parts; and their small size and modular nature makes them less exposed to dangers like earthquakes.

I'll have more to say about SMRs, other innovations in this layer, and some potential emerging investment opportunities in just a bit.

Overall, you could say the power generation layer is the core value-adding stage of the nuclear power value chain. It takes the prepared nuclear fuel and produces energy from it. But it still couldn't exist without the other layers in the chain.

This layer includes today's large plants and tomorrow's innovative reactors. It's highly capital-intensive—building reactors costs billions—but once they're up and running, plants generate massive amounts of electricity at a very low cost.

The support layer: Maintaining and servicing nuclear power plants

This is the "keeping everything running smoothly and safely" layer of the nuclear power value chain.

There's a whole ecosystem of experts and technologies necessary to keep nuclear power plants running safely and efficiently throughout their 60- to 80-year life cycle.

You can think of this ecosystem as a nuclear plant's pit crew and guardian angels. Some key support layer components include:

Maintenance and repair services

Once a plant is up and running, maintenance is key. Every 18–24 months or so, it's shut down for a few weeks for partial refueling and maintenance.

During these outages, teams might replace pumps, check valves and sensors, inspect welds for any signs of wear, and test safety systems. Even large parts like steam generators and reactor vessel heads get replaced over time.

There's an entire industry of companies that assist with all this.

For example, Westinghouse and Framatome not only design reactors but also provide refueling services and replacement parts to plants worldwide. And companies like **General Electric (GE)** provide turbine-generator maintenance.

Meanwhile, new techniques are being introduced to service reactors more efficiently. Robots and drones are now being used to inspect hard-to-reach areas and do things like look for cracks in reactor vessel welds and survey containment buildings.

Technical services and consulting

Because conventional nuclear plants are so complex, there's a need for engineering and scientific support throughout their lives.

Engineering companies like Bechtel, **Fluor (FLR)**, or **Jacobs Solutions (J)** might be contracted to design plant upgrades—like replacing analog control systems with digital ones or "uprating" a reactor to produce more power by improving cooling or fuel design.

There are also consulting firms and software providers needed to help optimize plant performance—like data analytics to predict when a pump might need maintenance or repair to save money and prevent downtime.

Training services are also required. Companies like GSE Solutions and **L3Harris Technologies (LHX)** provide simulator training technology and certification programs for reactor operators.

Safety systems and solutions

Modern reactors are built with multiple layers of safety to contain radiation—like the thick concrete and steel containment building around the reactor as well as a series of emergency backup systems. But they still need a lot of ongoing support from the safety side of things.

For example:

- Companies like FoxGuard Solutions and Dragos protect plants from cyber-attacks with cybersecurity systems specifically designed for nuclear facilities.
- Companies like **Mirion Technologies (MIR)** and **Thermo Fisher Scientific (TMO)** provide radiation detection, monitoring, and measurement systems.
- Companies like **Honeywell International (HON)** and **Johnson Controls International PLC (JCI)** provide fire protection and security systems.

- Companies like Jensen Hughes and **Tetra Tech (TTEK)** provide ongoing safety consulting and risk assessment.

- And companies like Platom and SECTOR provide specialized safety and compliance services.

That's not an exhaustive list by any means, but you get the idea.

By keeping nuclear power plants running smoothly and safely, the support layer adds and protects value. In a way, this layer is the "glue" holding the whole value chain together through the decades of a plant's operation. It requires much less capital than building reactors but is very knowledge- and labor-intensive.

The cleanup layer: Managing and disposing of nuclear waste

Fuel assemblies remain in the reactor contributing to power generation for several years. After that, the fuel is considered "spent" and becomes incredibly hot and highly radioactive waste that must be removed for storage or reprocessing/recycling.

Also, reactors don't last forever. When a plant reaches the end of its life, it must be safely decontaminated and dismantled (decommissioned).

So this layer is about managing the back end of the nuclear power cycle. It's expensive, but crucial for the environment and future generations.

Spent nuclear fuel is first stored in giant pools to cool for a few years. Then, it typically goes into massive concrete and steel casks and is stored underground at the site. This storage is temporary, but it's still designed to last for several decades.

Companies like **Perma-Fix Environmental Services (PESI)** and EnergySolutions specialize in treating all this nuclear waste and preparing it for storage.

Meanwhile, a Finnish company called Posiva is developing the world's first permanent nuclear waste storage repository.

Called ONKALO, it's the most impressive engineering project you've never heard of—a series of tunnels 400 meters underground in solid granite where nuclear waste should be safely stored for ***hundreds of thousands of years***.

Posiva's been actively developing ONKALO for over two decades. The company completed its first trial run in March of this year and expects full operational launch in 2026.

ONKALO is an exciting and absolutely necessary development for the nuclear power industry, as are similar repositories planned around the world. But nuclear waste recycling is more exciting.

The French multinational Orano operates facilities capable of recycling up to 96% of used nuclear fuel. This recycling process recovers uranium and plutonium to make new fuels like MOX (mixed oxide), which can run in certain reactors. It also reduces the volume and radiotoxicity of the waste by up to 5X and 10X, respectively.

Orano is bringing its recycling tech to countries like Japan, Germany, and the Netherlands. And the company recently teamed up with SHINE Technologies to develop a pilot recycling plant in the US, aiming to process 100 metric tons of used nuclear fuel annually by the early 2030s.

Basically, the cleanup layer ensures that the nuclear power value chain ends responsibly. It's about converting the value we got from the electricity into a form of liability management.

From a business perspective, this layer is considered a necessary cost rather than a profit center. But there are still opportunities here, as I alluded to above. For example, companies that develop superior waste storage technologies or more efficient recycling.

And if the advanced reactors being developed succeed in using spent fuel as new fuel, yesterday's waste becomes part of tomorrow's value chain foundation layer.

This isn't a "winner-takes-all" market

There's no shortage of companies working on advanced nuclear reactor technology.

Much of this research and development centers on shrinking the tech down to make it cheaper, safer, faster to build, easier to cool, and accessible to more places like remote regions or for commercial/industrial purposes—like powering AI data centers—in grid-strained areas.

These efforts have led to the development of SMRs—and their even smaller cousins called microreactors and nano-reactors—which aim to overcome the limitations of conventional large-scale nuclear power plants.

Publicly-traded companies developing these smaller advanced reactors include:

- **NuScale Power Corp. (SMR)**
- **Oklo (OKLO)**
- **BWX Technologies (BWXT)**
- **NANO Nuclear Energy (NNE)**
- **GE Vernova (GEV)**
- **Rolls-Royce Holdings PLC (RYCEY)**

The problem for investors: We're not going to see significant revenue from these efforts for several years still.

NuScale Power, for example, is the first and only SMR developer to receive US Nuclear Regulatory Commission (NRC) design approval. This is a major milestone in commercializing SMRs, but the company still doesn't expect to have one in operation until 2030. And that's the best-case scenario.

We already saw the cancelation of NuScale's Carbon Free Power Project to build SMRs at a site near Idaho Falls—which had been penciled in for operation by 2029—due to an enormous, expected cost increase.

The company does generate revenue from engineering and licensing fees, and from services provided to potential customers. But it's rather insignificant. Total revenue only reached $49 million in the past year. That's compared to a market cap of $5 billion.

In other words, NuScale is trading at a price-to-sales ratio of over 100 today. That's an extremely rich valuation, especially considering that it's hard to see revenue growing much in the near term before its tech is commercialized.

Meanwhile, management says the company is in advanced talks to build SMRs for five hyperscalers building AI data centers. But none of these discussions—or any others NuScale is having with potential customers—have resulted in a firm customer order yet.

I'm not saying NuScale is a bad company. It could end up being wildly successful and making early investors lots of money. I just think it's a little too early to invest in today, given the valuation, the uncertainty involved, and the near-term revenue prospects.

There are also lots of competition to consider.

On May 8, for example, GE Vernova's joint venture with Hitachi—called GE Vernova Hitachi Nuclear Energy, or GVH for short—announced that construction of its first SMR has been approved in Canada by the Province of Ontario and Ontario Power Generation.

GVH's press release about the approval reported that construction will begin soon, "with completion of the first unit scheduled by the end of the decade."

In other words, while GVH is beating NuScale outside of the US—and could become the first company in the West with an operational SMR—it's still about four years away at best.

NANO Nuclear Energy is an interesting company to watch because its microreactor tech shrinks things down even further than NuScale and GVH. The company's ZEUS and ODIN reactors are designed to fit on the back of a semitruck. So they're easily transportable, can be deployed to remote locations and military bases, and can be used in emergency response scenarios for reliable power.

NANO says its microreactors will be launched between 2030 and 2031, but there's a lot of uncertainty around that estimate.

Meanwhile, a bunch of private companies developing SMRs and microreactors—several of which Stephen recently met with in person—are nipping at the public companies' heels, too.

Here's a list of the most promising ones:

- **TerraPower** recently broke ground on its Natrium SMR project in Wyoming. It's expected to be operational around 2030. And it's particularly interesting because it can ramp up power fast like a gas-powered turbine while also offering energy storage for enhanced grid flexibility.

- **Radiant Nuclear** is aiming to commercialize its Kaleidos truck-portable microreactor tech by 2028.

- **Antares Industries** is developing microreactors with an emphasis on automation for niche applications like space power and remote military installations. A prototype demonstration unit is expected this year, but commercial deployment timelines remain unspecified.

- **Valar Atomics** plans to have test SMR deployments in 2026, with commercial "gigasites" designed for heavy industrial power in the early 2030s.

- **Last Energy's** plug-and-play reactor modules are designed to snap together like Legos. The company is aiming for commercial deployment in 2027 or 2028 in Poland, which means it could be first to the European market among private SMR developers.

- **Aalo Atomics'** extra modular reactors (XMRs) also take a Lego-block approach. They're tailored for the growing energy demands of AI and data centers, with the first full prototype slated for 2029.

- **Kairos Power** recently broke ground on its Hermes reactor, with commercial deployments expected to start in 2030. Importantly, Kairos signed a landmark agreement with Google last year to supply 500 MW of power from multiple SMRs by 2035.

The takeaway here for investors is that this isn't going to be a winner-takes-all market. We'll have SMRs, microreactors, and nano-reactors of all shapes and sizes supplied by different companies for different purposes. And there's going to be plenty of room for multiple winners.

But of all the companies developing this sort of advanced reactor technology, the one that looks like the best investment today is **BWX Technologies (BWXT)**.

We're not recommending BWXT today mainly because it doesn't meet our revenue growth requirements. But I want to give you a quick rundown of the company because that could change relatively soon.

BWX is a long-time critical nuclear equipment and fuel supplier for the US Navy's submarine and aircraft-carrier reactors. And its broad portfolio of nuclear components, specialty fuels, and engineering and manufacturing services is now giving investors a "picks-and-shovels" way to participate in the coming wave of SMR and microreactor deployment. It's already secured manufacturing work on GVH's and NuScale's SMRs, for example.

The company is also working on new microreactor tech. It's developing a mobile microreactor for the US Defense Department's Project Pele and its own advanced commercial microreactor.

BWX is also one of the only western companies already producing large quantities of the meltdown-proof TRISO nuclear fuel I talked about earlier.

Put simply, BWXT is the rare stock that offers meaningful exposure to the rapid growth in advanced reactor technology *and* an already profitable business to backstop the valuation if hiccups occur in the development and rollout of SMRs and microreactors.

America's fastest-growing stocks

In the table below, we share 25 of the fastest-growing stocks in America that fit our *Disruption_X* size criteria. As a reminder, the stocks we're targeting have a market cap between $200 million and $20 billion.

Future recommendations may come from this list. It also gives us great insight into what's growing fast and—just as important—what's not. A stock riding on pure hype can't make this list because it requires real, bona fide revenue growth.

Note that these stocks trade on American exchanges. They aren't necessarily American companies.

We also include their raw GARP Quotient based on their most recent quarterly revenue to give an idea of how fairly valued they are.

Observations this month:

Overall, the list shows the continued convergence of the physical and digital worlds. And it suggests that the future of disruption lies where sectors intersect. Innovations in one area like AI or quantum computing can accelerate progress in others like advanced tech for electric vehicles (EVs), driving multiple sectors forward simultaneously.

Speaking of EVs, the list suggests success is increasingly about luxury positioning and technology differentiation rather than just mass-market adoption. **ZEEKR Intelligent Technology Holding Limited (ZK)**, **Lotus Technology (LOT)**, and **Rivian Automotive (RIVN)** all focus on premium segments with advanced technology features.

The list also indicates that quantum computing is leaving the lab. **IonQ (IONQ)** is now guiding for up to $95 million in revenue this year. And **D-Wave Systems (QBTS)** reported year-over-year revenue growth of more than 500% in the first quarter thanks to the first sale of its Advantage quantum computer to a major research institution.

This shows the beginning of quantum's transition out of the lab to solving real-world problems in fields like drug discovery, materials science, and logistics.

<u>A few of the fast-growing stocks that have our attention</u>:

IonQ (IONQ) & **D-Wave Systems (QBTS)**—These two quantum computing stocks, as I alluded to above, are a lot more interesting now that they're generating real revenue from their technology.

Intuitive Machines (LUNR)—This space stock successfully landed its Odysseus lunar lander on the moon last year. It was the first ever moon landing by a commercial company. The company has also secured a $4.8 billion NASA contract for lunar infrastructure development.

Aeva Technologies (AEVA)—This sensing technology stock develops LiDAR and perception tech for autonomous systems. Think robotics, industrial automation, and next-gen vehicles. It sits at the intersection of sensing, computing, and physical-world disruption.

Fastest-growing stocks in America

Rank	Ticker	Name	Description	Market Cap ($M)	Stock Price	Revenue Growth	GARP Quotient
1	IMMR	Immersion Corporation	Haptic software	$254	$7.84	3539%	0.00
2	HYLN	Hyliion Corp.	Electric truck parts	$244	$1.39	807%	15.43
3	EH	EHang Ltd.	Chinese drones	$860	$16.43	288%	0.45
4	QBTS	D-Wave Quantum Inc.	Quantum technology	$4,680	$14.97	122%	64.14
5	ADSE	ADS-TEC Energy PLC	Battery systems	$718	$13.10	268%	2.18
6	LOT	Lotus Technology Inc. ADR	Luxury EV maker	$1,640	$2.42	20%	7.44
7	ODD	Oddity Tech Ltd.	AI beauty tech	$4,120	$73.78	114%	3.38

Fastest-growing stocks in America							
Rank	**Ticker**	**Name**	**Description**	**Market Cap ($M)**	**Stock Price**	**Revenue Growth**	**GARP Quotient**
8	CRDO	Credo Technology Group	Data infrastructure	$15,610	$91.92	126%	18.17
9	ZK	ZEEKR Intelligent Technology	Chinese EV	$6,710	$26.50	43%	0.18
10	MAX	MediaAlpha, Inc.	Ad tech platform	$733	$10.86	149%	0.47
11	CLSK	CleanSpark, Inc.	Bitcoin miner	$2,820	$10.04	89%	4.35
12	AMTM	Amentum, Inc.	Defense tech contractor	$5,420	$22.29	121%	0.32
13	INOD	Innodata Inc.	AI data services	$1,490	$46.96	114%	5.59
14	ACTG	Acacia Research Corporation	Patent licensing firm	$351	$3.65	65%	1.08
15	IONQ	IonQ, Inc.	Quantum technology	$10,690	$40.86	70%	505.18
16	SOUN	SoundHound AI, Inc.	Voice AI company	$3,950	$9.82	101%	33.44
17	LCID	Lucid Group, Inc.	American EV	$6,620	$2.17	41%	17.31
18	RIVN	Rivian Automotive, Inc.	American EV	$15,830	$13.81	1%	550.26
19	AEVA	Aeva Technologies, Inc.	Lidar sensors	$1,650	$29.97	96%	127.71
20	BYRN	Byrna Technologies Inc.	Aerospace & Defense	$741	$32.67	87%	8.10
21	ATEX	Anterix Inc.	Critical communications	$535	$28.78	67%	127.89
22	LUNR	Intuitive Machines, Inc.	Aerospace & Defense	$1,850	$10.34	62%	11.96
23	FTAI	FTAI Aviation Ltd.	Aircraft leasing	$13,850	$135.05	58%	11.92
24	ATAT	Atour Lifestyle Ltd.	Chinese hotel chain	$3,680	$32.40	55%	0.80
25	VFS	VinFast Auto Ltd.	Vietnamese EV	$8,100	$3.46	59%	0.00

For the most up-to-date guidance on our current positions, click here to view the portfolio online.

Chris Wood Stephen McBride

Chris Wood and Stephen McBride

Keep up with RiskHedge on the go.

Download the App

Scan it with your Phone

READ IMPORTANT DISCLOSURES HERE.

YOUR USE OF THESE MATERIALS IS SUBJECT TO THE TERMS OF THESE DISCLOSURES.